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SEARCHING ALGORITHMS

Performance Comparison Table

Algorithm	Strategy	Data Structure	Steps Taken	Optimal Solution	Memory Usage
BFS	Level-by-level search	Queue	5	Yes	High
DFS	Depth-first exploration	Stack	5	Not guaranteed	Low
Iterative DFS	DFS using controlled stack	Stack	7	Not guaranteed	Low
DLS	DFS with depth limit	Recursion	7+	Not guaranteed	Medium
UCS	Expands lowest cost node	Priority Queue	5	Yes	High

Detailed Comparison

Feature	BFS	DFS	Iterative DFS	DLS	UCS
Search Type	Uninformed	Uninformed	Uninformed	Uninformed	Uninformed
Completeness	Yes	No	Yes (with limit)	Depends on limit	Yes
Optimality	Yes	No	No	No	Yes
Time Complexity	O(b^d)	O(b^m)	O(b^d)	O(b^l)	O(b^d)
Space Complexity	O(b^d)	O(bm)	O(bm)	O(bl)	O(b^d)

Where:

- **b** = branching factor
- **d** = depth of solution
- **m** = maximum depth
- **l** = depth limit

Observations

1. **BFS and UCS produced the shortest path** to the goal.
2. **DFS also reached the goal**, but DFS does not always guarantee optimal solutions.
3. **Iterative DFS explored deeper paths first**, resulting in more steps.
4. **DLS repeated some states due to recursive exploration**.
5. **UCS behaves similar to BFS here because all actions have equal cost.**

Conclusion

The Milk Jug problem was successfully solved using multiple **Uninformed Search Algorithms**.

- **BFS and UCS** provided the most optimal solutions.
- **DFS variants** required more exploration but used less memory.
- The experiment demonstrates how different search strategies affect **efficiency, memory usage, and optimality**.

These algorithms form the foundation of **Artificial Intelligence problem-solving techniques** used in robotics, game solving, and planning systems.